



Program	BACHELOR OF TECHNOLOGY (B.Tech)	Semester - 5
Type of Course	Professional Core Course	
Prerequisite		
Rationale	-	
Effective From A.Y.	2026-27	

Teaching Scheme (Contact Hours)				Examination Scheme				
Lecture	Tutorial	Lab	Credit	Theory Marks		Practical Marks		Total Marks
				SEE TH	IAT	SEE P	CCE	
3	0	2	4	70	30	50	50	200

SEE - Semester End Examination, IAT - Internal Assessment Test, CCE - Continues & Comprehensive Evaluation

Course Content		T - Teaching Hours W - Weightage	
Sr.	Topics	T	W
1	Introduction to AI and Intelligent Agents What is Artificial Intelligence: definitions, goals, Turing test, Brief history and milestones, Types of AI, Subfields of AI, Applications of AI in IT and industry, Intelligent agents and environments, Agent types, PEAS representation and performance measures, Properties of environments, Problem formulation for AI	5	10
2	Problem Solving by Search State-space search and search concepts, Uninformed search strategies, Informed (heuristic) search, Uniform-cost Search, Discussion of completeness, optimality, time and space complexity (intuitive level), admissible and consistent heuristics, heuristic design and evaluation, Local search and optimization, Simulated annealing and Genetic algorithms	9	20
3	Constraint Satisfaction and Adversarial Search Constraint Satisfaction Problems (CSPs), Solving CSPs with backtracking search, Improvements using forward checking and basic constraint propagation, Games and adversarial search: game trees, types of games, Minimax algorithm for two-player deterministic games, alpha-beta pruning and its effect on search efficiency, Introduction to planning, STRIPS-like representation	9	20
4	Knowledge Representation and Propositional Logic Knowledge-based agents and the role of knowledge in AI systems, Knowledge representation issues, Declarative vs Procedural knowledge Propositional logic: syntax and semantics, logical equivalence, validity and satisfiability Inference in propositional logic: inference rules, soundness and completeness, resolution in propositional logic with simple examples.	4	10
5	First-Order Logic and Inference Motivation for First-Order Logic (FOL) compared to propositional logic, Syntax and semantics of FOL, Translating natural language statements into FOL, Inference in FOL, Resolution in FOL Overview of other knowledge representation schemes: Semantic networks, frames, production rules and basic expert systems	9	20
6	Uncertainty and Probabilistic Reasoning	9	20

**Course Content**

T - Teaching Hours | W - Weightage

Sr.	Topics	T	W
	Limitations of purely logical reasoning in real-world domains; motivation for probabilistic models, Basics of probability theory relevant to AI, Bayesian reasoning for simple diagnostic problems, Introduction to Bayesian networks Machine Learning and Modern AI : Relationship between AI and ML, Data-driven vs knowledge-driven approaches, Overview of supervised learning, Overview of unsupervised learning, Introduction to neural networks and deep learning, Modern AI systems and large language models (LLMs), Examples of LLM-based applications, capabilities, limitations and ethical/societal issues.		
Total		45	100

Suggested Distribution Of Theory Marks Using Bloom's Taxonomy

Level	Remembrance	Understanding	Application	Analyze	Evaluate	Create
Weightage	20	30	25	25	0	0

NOTE : This specification table shall be treated as a general guideline for the students and the teachers. The actual distribution of marks in the question paper may vary slightly from above table.

Course Outcomes**At the end of this course, students will be able to:**

C01	Explain AI goals, history, types, applications and intelligent agents, and formulate simple IT problems as state space models.
C02	Apply uninformed and informed search algorithms to solve state space problems and compare their performance.
C03	Apply CSP and game search techniques to small constrained problems.
C04	Apply propositional logic representations to derive conclusions from knowledge bases.
C05	Analyse problem domains using first order logic, performing translation, unification and rule based inference.
C06	Analyze AI tasks using basic probability, Bayesian reasoning, Simple ML models, Modern AI/LLMs. Analyze AI tasks using basic probability, Bayesian reasoning, Simple ML models, Modern AI/LLMs.

**CO PO Mapping**

CO	CO - 1	CO - 2	CO - 3	CO - 4	CO - 5	CO - 6
PO - 1						
PO - 2						
PO - 3						
PO - 4						
PO - 5						
PO - 6						
PO - 7						
PO - 8						
PO - 9						
PO - 10						
PO - 11						

Reference Books

1.	Introduction to Artificial Intelligence and Expert Systems By Dan W. Patterson, PHI
2.	Artificial intelligence: A Modern Approach (TextBook) By Stuart Russell, Peter Norvig Prentice Hall. 4, Pub. Year 2020
3.	Artificial Intelligence (TextBook) By Elaine Rich, Kevin Knight, Shivashankar B. Nair Tata McGraw Hil 3rd
4.	Fundamentals of Artificial Intelligence By K. R. Chowdhary Springer/PHI
5.	Machine Learning By Tom Mitchell McGraw Hill, Pub. Year 2017
6.	Artificial Intelligence: A New Synthesis By Nils J. Nilsson Morgan Kaufmann

**List of Practical**

1.	Python environment setup; representing AI problems as graphs/state spaces Aim: To set up the Python environment and represent simple AI problems using appropriate data structures such as graphs and state space models.
2.	Implementation of BFS and DFS for graph / maze problems Aim: To implement Breadth First Search and Depth First Search algorithms in Python and apply them to solve graph and maze traversal problems.
3.	Uniform cost search and A* search with heuristic design and comparison Aim: To implement Uniform Cost Search and A* search, design simple heuristic functions, and compare their performance with uninformed search strategies.
4.	CSP: backtracking and forward checking for map coloring / N queens Aim: To model constraint satisfaction problems and implement backtracking with forward checking to solve map coloring or N queens efficiently.
5.	Game playing: minimax and alpha-beta pruning for a two player deterministic game Aim: To implement minimax and alpha-beta pruning for a two player deterministic game and study their effect on search efficiency.
6.	Propositional logic: knowledge base representation and inference using rules/resolution Aim: To represent knowledge in propositional logic and perform automated inference using rules and resolution on a small knowledge base.
7.	First-order logic: domain modeling, unification and simple rule-based inference Aim: To model a domain in first order logic, apply unification, and perform simple rule based inference using forward or backward chaining.
8.	Probabilistic reasoning: Bayes' rule computations and naive Bayes classifier implementation Aim: To apply Bayes' rule for probabilistic reasoning and implement a naive Bayes classifier for a basic classification task.
9.	Using AI tools for AI problem solving and code development Aim: To use modern AI tools to assist in problem understanding, algorithm design, code generation, debugging, and documentation for AI programs, and to critically evaluate their benefits, limitations, and ethical issues.
10.	Mini project: Small AI application Aim: To design and implement a small AI application that integrates suitable techniques and document its problem formulation, methodology and results.